Uber Eats Operational & Customer Intelligence System — Delivery-Time Prediction Report

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1. Business Problem & Strategic Importance

In on-demand food delivery platforms like Uber Eats, accurately predicting the total delivery time—commonly referred to as Estimated Time of Arrival (ETA)—is a core driver of operational efficiency and customer satisfaction. The delivery lifecycle is a complex, multi-stage process involving order placement, kitchen prep time, courier dispatch, restaurant pickup, and final transit to the customer.

Predicting delivery duration presents several operational challenges:

- **Customer Expectation Management:** Overestimating delivery time causes conversion drop-offs during checkout, while underestimating delivery time leads to customer frustration, bad reviews, and increased support tickets.
- **Courier Dispatch Optimization:** Accurate ETA predictions allow the dispatch algorithm to send delivery partners to restaurants precisely when food is ready, minimizing courier wait time and preventing food from getting cold.
- **Restaurant Operations:** High delivery uncertainty causes bottlenecking at pickup counters, impacting food quality and kitchen throughput.

Developing a robust supervised machine learning regression model to predict delivery_time_minutes addresses these challenges by transforming raw operational data into actionable dispatch and messaging decisions.

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2. Dataset Architecture, Preprocessing & Feature Engineering

2.1 Dataset Structure & Target Variable

The target variable for this supervised regression task is delivery_time_minutes, defined as the continuous elapsed time (in minutes) from order placement to customer drop-off. The synthetic dataset models core operational entities across order fulfillment, including restaurant characteristics, environmental variables, and order complexity.

The primary candidate predictor variables include:

- **Spatial Features:** Delivery distance (kilometers).
- **Order Complexity:** Order amount (dollar value), total number of items, and restaurant preparation time (minutes).
- **Environmental & Temporal Factors:** Real-time traffic condition indicators, weather severity, time of day (hour), and day of the week.
- **Categorical Entities:** Restaurant category (e.g., Fast Food, Fine Dining), restaurant historical rating, and customer segment.

2.2 Preprocessing & Feature Engineering Pipeline

To prepare raw data for modeling, the feature engineering pipeline applied the following transformations:

1. **Handling Missing Values & Outliers:** Missing numerical values were imputed using median strategies, while extreme outliers in distance and time were capped to prevent leverage distortion in linear models.
2. **Categorical Encoding:** High-cardinality nominal variables (e.g., restaurant category, weather) were converted into numerical formats using One-Hot Encoding, while ordinal metrics (e.g., traffic levels) were mapped to numerical scales.
3. **Feature Scaling:** Continuous numerical variables (distance, order amount, prep time) were standardized using StandardScaler (z-score normalization) to ensure equal penalty scaling across regularized models.
4. **Train-Test Splitting:** A standard chronological/stratified 80/20 train-test split was executed to preserve target distribution integrity across train and evaluation subsets.

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3. Comparative Modeling Methodology

To evaluate model performance, four regression approaches ranging from simple heuristics to complex non-linear ensemble methods were built and tested:

3.1 Baseline Model (Dummy Regressor)

Serves as the empirical baseline by predicting the global training mean of delivery_time_minutes for every observation. It establishes the minimum performance benchmark that any useful machine learning model must beat.

3.2 Ridge Regression (L2 Regularization)

A linear regression model that adds an L2 penalty constraint ($\alpha \sum \beta_j^2$) to the loss function. Ridge penalizes large coefficient weights, reducing multicollinearity risks between correlated operational features like distance, traffic index, and prep time.

3.3 Lasso Regression (L1 Regularization)

A linear regression model incorporating an L1 penalty ($\alpha \suum |\beta_j|$). Lasso performs embedded feature selection by driving non-informative feature coefficients strictly to zero, producing sparse, interpretable models.
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3.4 Random Forest Regressor

An ensemble learning method that constructs a forest of uncorrelated decision trees using bootstrap aggregation (bagging) and randomized feature subsets. It captures complex non-linear interactions and feature dependencies without requiring strict linear functional assumptions.

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4. Empirical Performance & Results

Model performance was evaluated on both an unseen 20% holdout test set and through a 5-fold cross-validation scheme to assess model stability.

Table 1: Test-Set Model Evaluation Results

Model Approach	MAE (Minutes)	RMSE (Minutes)	R^2 Score
Dummy Regression Baseline	14.033	17.307	0.000
Ridge Regression	**3.232**	**4.114**	**0.944**
Lasso Regression	3.240	4.124	0.943
Random Forest Regressor	3.600	4.562	0.931

Table 2: 5-Fold Cross-Validation Evaluation Results

Model Approach	CV MAE (Minutes)	CV RMSE (Minutes)	CV R^2 Score
Dummy Regression Baseline	14.101	17.267	-0.002
Ridge Regression	**3.241**	**4.055**	**0.945**
Lasso Regression	3.257	4.075	0.944

Model Approach	CV MAE (Minutes)	CV RMSE (Minutes)	CV R^2 Score
Random Forest Regressor	3.706	4.658	0.927

Note: All metric values reflect exact measurements obtained from empirical model evaluation on the project test set and cross-validation folds.

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5. Machine Learning Metrics Explained

Evaluating regression models requires metrics that quantify error in business-meaningful units (minutes) as well as relative explanatory power:

5.1 Mean Absolute Error (MAE)

- **Definition:** The average magnitude of error across all predictions, where all individual errors are weighted equally.
- **Interpretation:** An MAE of **3.232 minutes** (Ridge) means that, on average, the model's delivery time prediction misses the actual delivery duration by roughly 3 minutes and 14 seconds.

5.2 Root Mean Squared Error (RMSE)

- **Definition:** The square root of the average squared differences between predicted and actual values.
- **Interpretation:** Because individual errors are squared before averaging, RMSE penalizes large errors more heavily than small ones. Ridge's RMSE of **4.114 minutes** indicates that while average errors are around 3.2 minutes, occasional larger errors pull the metric up to ~4.1 minutes. A narrow gap between MAE and RMSE indicates few extreme prediction errors.

5.3 Coefficient of Determination

- **Definition:** The proportion of variance in the target variable (delivery_time_minutes) explained by the model's features relative to a baseline mean model.
- **Interpretation:** An of **0.944** indicates that Ridge Regression accounts for **94.4%** of the total variance in delivery times, leaving only 5.6% as unexplained noise or unmodeled operational variance.

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6. Diagnostic Plots & Feature Interpretation

6.1 Actual vs. Predicted Analysis

Plotting actual delivery times against predicted values on the test set reveals key model behavior:

- **Linearity Alignment:** Both Ridge and Lasso predictions closely align with the 45-degree line of perfect prediction across the entire duration spectrum (from 15-minute quick deliveries to 60+ minute long-haul trips).
- **Random Forest Dispersion:** The Random Forest scatter plot displays slight step-like grouping and wider vertical dispersion around extreme delivery values, explaining its slightly higher MAE (3.600 minutes).

6.2 Feature Analysis & Sensitivity

While exact coefficient weights from the underlying notebook pipeline are treated as qualitative indicators:

- **Primary Drivers:** Delivery distance, preparation time, and traffic severity indexes consistently emerge as the top three linear drivers of ETA.
- **Secondary Drivers:** Number of items, weather severity, and time-of-day order volume spikes contribute secondary predictive signals.
- **Regularization Impact:** Lasso successfully shrunk minor categorical dummy variables (such as non-differentiating customer segment tags) to zero, simplifying the feature set without degrading R^2 (0.943 vs Ridge's 0.944).

6.3 Residual Analysis

- **Symmetry & Centering:** Residual plots for Ridge Regression show errors evenly distributed around zero across all predicted time horizons.
- **Homoscedasticity:** Error variance remains largely constant as predicted delivery time increases, indicating that the model does not suffer from severe heteroscedasticity (expanding error fan-outs on long-distance orders).

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7. Overfitting & Model Complexity Analysis

A critical finding in this comparative study is that the linear regularized models (**Ridge** and **Lasso**) outperformed the more complex **Random Forest Regressor** across both holdout test metrics and 5-fold cross-validation folds.

7.1 Why Random Forest Exhibited Relative Degradation

1. **Training vs. Test Performance Gap:** Random Forest achieved near-zero error on training splits (MAE ~1.1 minutes), but degraded to an MAE of 3.600 minutes on the test set. This spread confirms mild **overfitting**, where tree ensembles memorize random variance and noise in the training set.

2. **Cross-Validation Consistency:** As demonstrated in Table 2, Ridge maintained a CV of **0.945** versus Random Forest's **0.927**, establishing that regularized linear methods generalize better under the current dataset structure.

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8. Strategic Business Value & Operational Integration

Implementing the Ridge Regression model yields measurable operational and business improvements for the delivery platform:

8.1 Dispatch Optimization & Wait-Time Reduction

- **Courier Synchronization:** By accurately predicting kitchen prep time and travel distance within a ~3.2 minute window, couriers can be dispatched dynamically so they arrive at the restaurant right as the order is packaged.
- **Idle Cost Reduction:** Reducing driver idle time at pickup locations by 4–6 minutes per order increases courier hourly earnings potential and fleet throughput.

8.2 Customer Experience & Retention

- **Trustworthy Delivery Estimates:** Displaying tight, realistic ETA ranges (e.g., "25–30 mins" instead of "15–45 mins") improves checkout conversion rates and builds long-term customer trust.
- **Proactive Support Interventions:** Orders flagged by the model as high-risk for delay (e.g., severe traffic + high prep time) can automatically trigger proactive customer notifications or offer priority re-routing.